

1.

$$\int \frac{1}{x + \sqrt{x}} dx$$

Answer: $2 \ln | \sqrt{x} + 1 | + C$

Solution: Take a u-substitution $x = u^2$ to get:

$$\int \frac{2u}{u^2 + u} du = \int \frac{2}{u + 1} du = 2 \ln | u + 1 | + C = 2 \ln | \sqrt{x} + 1 | + C$$

2.

$$\int_0^\infty \frac{4x}{1 + x^4} dx$$

Answer: π

Solution: Do a u-sub with $u = x^2$.

3.

$$\int_1^\infty \frac{1}{x^2 \sqrt{x^2 - 1}} dx$$

Answer: 1

Solution: trig sub

4.

$$\int_1^\infty \ln \left(1 + \frac{1}{x} \right) - \frac{1}{x + 1} dx$$

Answer: $1 - \ln(2)$.

Solution: Integration by parts by deriving $\ln(1 + \frac{1}{x})$ as a whole. Then

$$\left[x \ln \left(1 + \frac{1}{x} \right) \right]_1^\infty = 1 - \ln(2)$$

5.

$$\int \frac{\sin^4 \theta}{\cos \theta} d\theta$$

Answer: $-\frac{\sin^3 \theta}{3} - \sin \theta + \ln | \sec \theta + \tan \theta | + C$

Solution: Simplify using trig identities:

$$\begin{aligned} \int \frac{(1 - \cos^2 \theta)^2}{\cos \theta} d\theta &= \int \cos^3 \theta - 2 \cos \theta + \sec \theta d\theta \\ &= \int \cos \theta (1 - \sin^2 \theta) - 2 \cos \theta + \sec \theta d\theta \\ &= \int -\sin^2 \theta \cos \theta - \cos \theta + \sec \theta d\theta \\ &= -\frac{\sin^3 \theta}{3} - \sin \theta + \ln | \sec \theta + \tan \theta | + C \end{aligned}$$

6.

$$\int_0^\infty 2 - \frac{x}{\sqrt{x^2 + \pi^2}} - \frac{x}{\sqrt{x^2 + e^2}} dx$$

Answer: $e + \pi$.

Solution: Basic integration gives

$$= \left[2x - \sqrt{x^2 + \pi^2} - \sqrt{x^2 + e^2} \right]_0^\infty$$

Note that for any real value a :

$$\lim_{x \rightarrow \infty} (x - \sqrt{x^2 + a^2}) = 0.$$

7.

$$\int_{-1}^8 \sqrt{x} \sin(\tan^{-1}(\sqrt{x})) dx$$

Answer: 12

Solution: the integrand becomes $\frac{x}{\sqrt{x+1}}$

8.

$$\int \frac{\sin 2x}{1 + \cos x} dx$$

Answer: $2(\ln|\csc x + \cot x| - \ln|\csc x| - \cos x) + C$

Solution: Use trig rules and simplify/integrate.

9.

$$\int \frac{x+1}{x^2+x \ln x} dx$$

Answer: $\ln(x + \ln x)$

Solution: u-sub

10.

$$\int_0^{\frac{3}{2}} \sqrt{1 + \sqrt{x^2 + \sqrt{x^6 + \sqrt{x^{14} + \sqrt{x^{30} + \dots}}}}} dx$$

Answer: $\frac{3}{2} + \ln(2)$.

Solution: Notice that $y = \sqrt{1 + xy}$. Then $\frac{y^2-1}{y} = x$. By inverse sum formula,

$$\int_0^{3/2} f(x) dx + \int_1^2 f^{-1}(x) dx = 3$$

Hence,

$$I = 3 - \int_1^2 x - \frac{1}{x} dx = \boxed{\frac{3}{2} + \ln(2)}$$

11.

$$\int_0^{3/2} \frac{1}{x + \frac{1}{x+\dots}} dx$$

Answer: $16 \ln 2 + 3$

Solution: Let $u = \frac{1}{x + \frac{1}{x + \dots}}$, solve for u and x in terms of the other, and do a u-sub.

12.
$$\int_0^\pi \frac{6 + 6 \cos^2 x}{1 + 3 \sin^2 x} dx$$

Answer: 5π

Solution: Trig

13.
$$\int_0^1 \sqrt{x + x^3} + \sqrt{x + \sqrt[3]{x}} dx$$

Answer: $\boxed{\frac{4\sqrt{2}}{3}}$.

Solution: On the second term, let $u^3 = x$. Then we have

$$\int_0^1 \sqrt{x + \sqrt[3]{x}} dx = \int_0^1 3u^2 \sqrt{u^3 + u} du.$$

Then it follows

$$\int_0^1 \sqrt{x + x^3} + \sqrt{x + \sqrt[3]{x}} dx = \int_0^1 (3x^2 + 1) \sqrt{x + x^3} dx = \left[\frac{2}{3} (x + x^3)^{\frac{3}{2}} \right]_0^1 = \frac{4\sqrt{2}}{3}.$$

14.
$$\int_0^{\frac{\pi}{4}} \sum_{n=2}^{\infty} \left(\frac{1}{\sqrt{n - \sec^2(x)}} - \frac{1}{\sqrt{n - \tan^2(x)}} \right) dx$$

Answer: $\boxed{\frac{\pi}{2\sqrt{2}}}$.

Solution: Using the trig-identitiy makes the sum telescope to $\frac{1}{\sqrt{1 - \tan^2(x)}}$.

$$\begin{aligned} &= \int_0^{\frac{\pi}{4}} \frac{dx}{\sqrt{1 - \tan^2(x)}} = \int_0^1 \frac{du}{(1 + u^2)\sqrt{1 - u^2}} = \int_0^{\frac{\pi}{2}} \frac{d\theta}{1 + \sin^2 \theta} \\ &= \int_0^{\frac{\pi}{2}} \frac{\sec^2 \theta}{1 + 2 \tan^2 \theta} d\theta = \frac{\pi}{2\sqrt{2}} \end{aligned}$$

15.
$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{4 \log^2(\csc^2 x)}{\tan x} dx$$

Answer: $\log^2(2)$

Solution: u-sub with $u = \sin^2 x$

16.
$$\int e^{17x} (\cos^3 x - 3 \sin^2 x \cos x) dx$$

Answer: $\frac{e^{17x}(17 \cos(3x) + 3 \sin(3x))}{298} + C$

Solution: The trigonometric part is $\cos(3x)$, so one can use Euler's Identity and take the real part of the integral, or just do integration by parts.

17.
$$\int e^{x \tan^{-1}(x)} \frac{\tan^{-1}(x)}{\sqrt{x^2 + 1}} dx$$

Answer: $e^{x \tan^{-1}(x) - \ln(\sqrt{x^2 + 1})} + C$

Solution: Very sneaky to see, but

$$\frac{1}{\sqrt{x^2 + 1}} = e^{-\ln(\sqrt{x^2 + 1})}.$$

Then let $u = x \tan^{-1}(x) - \frac{1}{2} \ln(x^2 + 1)$ to easily solve the integral.

18.
$$\int_0^\varphi e^{x - \frac{1}{x}} (x + 1)^2 dx$$

Answer: $\varphi^2 e$.

Solution:

$$\begin{aligned} &= \int_0^\varphi \left(1 + \frac{1}{x^2}\right) e^{x - \frac{1}{x}} \cdot x^2 + 2x \cdot e^{x - \frac{1}{x}} dx \\ &= \left[x^2 e^{x - \frac{1}{x}} \right]_0^\varphi = \varphi^2 e \end{aligned}$$

19.
$$\int \frac{1}{x^5 + x^3} - \frac{3 \tan^{-1}(x)}{x^4} dx$$

Answer: $\frac{\tan^{-1}(x)}{x^3} + C$

Solution: It's best to go backwards with this one. As complicated as this looks, it's actually a quotient rule in disgused. The x^{-3} in the first term should be suggestive, along with $x^2 + 1$ term. We compute the derivative of $\frac{\tan^{-1}(x)}{x^3}$ and go backwards:

$$\begin{aligned} \frac{\frac{x^3}{x^2 + 1} - 3x^2 \tan^{-1}(x)}{x^6} &= \frac{x^3 - 3x^2(x^2 + 1) \tan^{-1}(x)}{x^6(x^2 + 1)} \\ &= \frac{1}{x^5 + x^3} - \frac{3 \tan^{-1}(x)}{x^4} \end{aligned}$$

20.
$$\int_0^{\frac{\pi}{4}} \frac{1}{\sqrt[4]{2} + \sqrt{1 + \tan(x)}} dx$$

Answer: $\frac{\pi}{8\sqrt[4]{2}}$

Solution: Secretly a King's rule problem, even if it doesn't look it. Let $u = \frac{\pi}{4} - x$.

$$\begin{aligned}
I &= \int_0^{\frac{\pi}{4}} \frac{1}{\sqrt[4]{2} + \sqrt{1 + \tan(\frac{\pi}{4} - u)}} du \\
&= \int_0^{\frac{\pi}{4}} \frac{1}{\sqrt[4]{2} + \sqrt{1 + \frac{1 - \tan(u)}{1 + \tan(u)}}} du \\
&= \int_0^{\frac{\pi}{4}} \frac{1}{\sqrt[4]{2} + \sqrt{\frac{2}{1 + \tan(u)}}} du \\
&= \int_0^{\frac{\pi}{4}} \frac{\frac{\sqrt{1 + \tan(u)}}{\sqrt[4]{2}}}{\sqrt{1 + \tan(u)} + \sqrt[4]{2}} du.
\end{aligned}$$

Hence,

$$2I = \int_0^{\frac{\pi}{4}} \frac{1 + \frac{\sqrt{1 + \tan(x)}}{\sqrt[4]{2}}}{\sqrt{1 + \tan(x)} + \sqrt[4]{2}} dx = \int_0^{\frac{\pi}{4}} \frac{1}{\sqrt[4]{2}} dx = \frac{\pi}{4\sqrt[4]{2}}$$

and the answer is $\boxed{\frac{\pi}{8\sqrt[4]{2}}}$.